

Problem Solving 2

```
int bar(x, y)
```

```
| if (y == 0) return 0
```

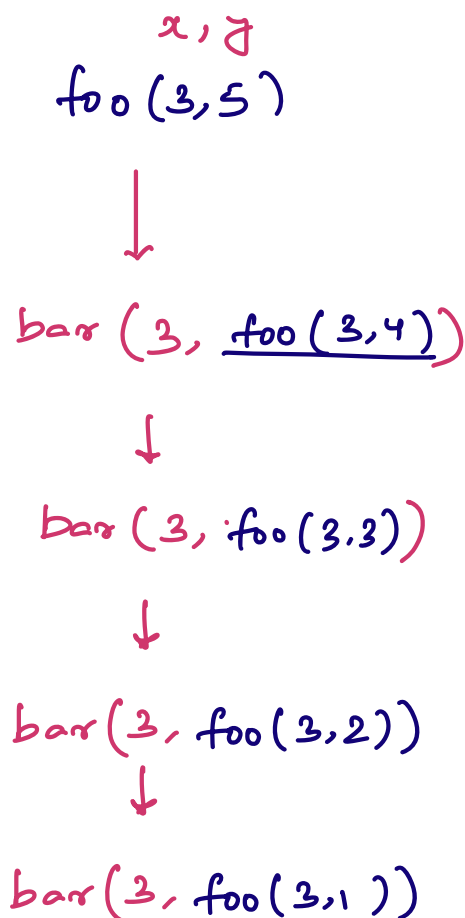
```
| return x + bar(x, y-1);
```

foo(3, 5)

```
int foo(x, y)
```

```
| if (y == 0) return 1
```

```
| return bar(x, foo(x, y-1));
```



$$\begin{array}{c} \downarrow \\ \text{bar}(3, \overset{1}{\text{foo}(3,0)}) \quad (3) \\ \swarrow \\ 3 + \text{bar}(3,0) \\ \quad \quad \quad 0 \end{array}$$

Is magic?

$$\begin{array}{c} \nearrow \text{magic} \\ 93421 \Rightarrow 19 \Rightarrow 10 \Rightarrow 1 \quad \underline{\underline{\text{true}}} \end{array}$$

$$83557 = 28 = 10 = 1 \quad \underline{\underline{\text{true}}}$$

* Divisibility rule of 9

↳ sum of all the digits should be multiple of 9

$$\begin{array}{c} \underline{237} \rightarrow 12 \rightarrow \boxed{3} \rightarrow \text{False} \\ 234 + 3 \end{array}$$

$$235 \rightarrow 10 \rightarrow \boxed{1}$$

$$\begin{array}{r} 5 \\ 26 \\ 9 \overline{) 237} \\ \underline{18} \\ 57 \\ \underline{54} \\ 3 \end{array}$$

Way 1

```
int magicno (int A)
{
    if (A < 10) {
        if (A == 1) return 1;
        else return 0;
    }

    int sum = 0;
    while (A > 0) {
        sum += A % 10;
        A = A / 10;
    }

    return magicno (sum);
}
```

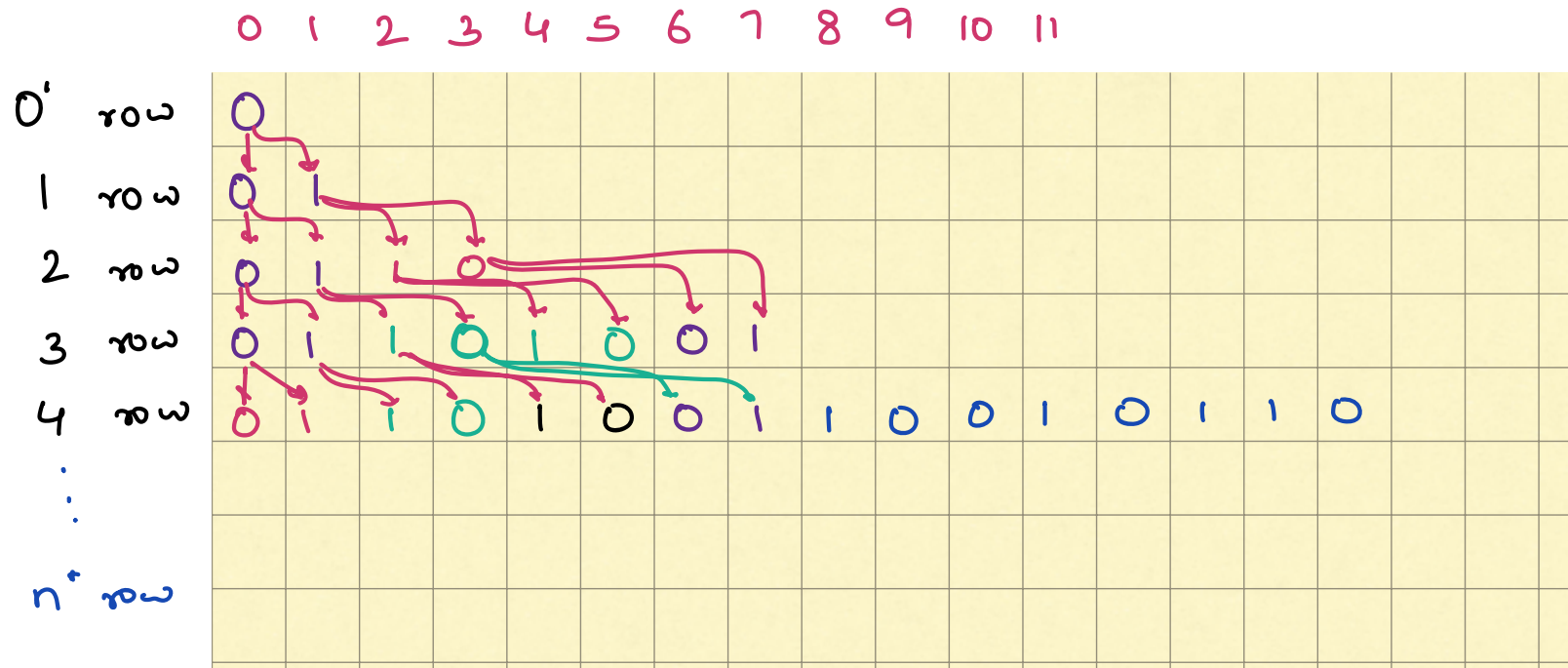
Way 2

```
int magic (int A) {
    if (A % 9 == 1)
        return 1;
    else return 0;
}
```

K^{th} symbol

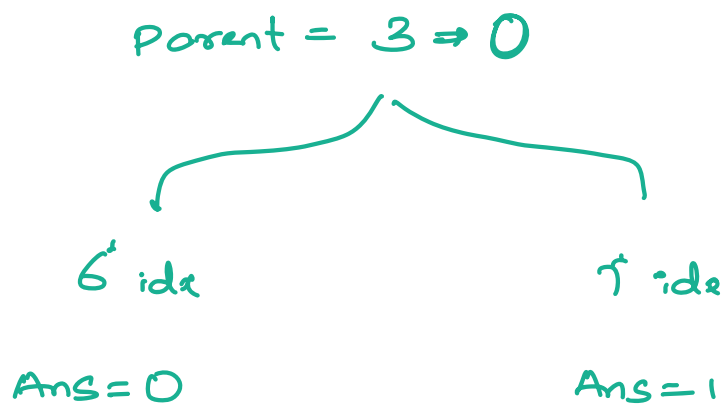
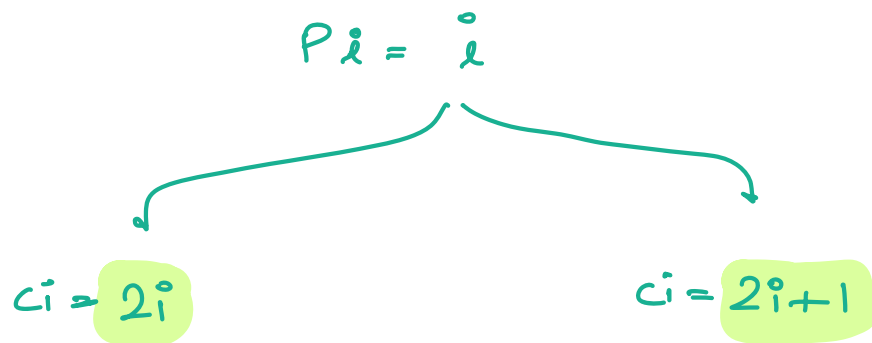
Given N, K . Find the K^{th} ele in N^{th} row.

$0 \rightarrow \{0, 1\}$ $1 \rightarrow \{1, 0\}$



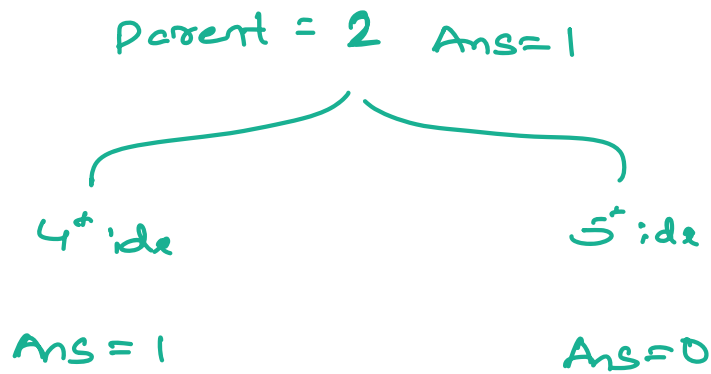
N	K		
2	3	$\rightarrow$	0
3	4	$\rightarrow$	1
4	8	$\rightarrow$	1

$p_i =$	c_i
2	4 th idx, 5 th idx
4	8 th idx, 9 th idx
3	6 th idx, 7 th idx



Obs $\rightarrow$ Ans for c_i will be same as parent if c_i is even child of parent

$\rightarrow$ if it is odd child $\rightarrow$ ans will be differ



```
int k'symbol (A, B)
```

```
if (A == 0 || B == 0) return 0;
```

TC: $O(\log n)$

```
int p = k'symbol(A-1, B/2)
```

→ Asking par for its answer

```
if (B % 2 == 0) return p;
```

```
else return 1-p
```

} generate your ans from subproblem

* change character

String A of size N, value = B

You can change atmost B characters, $\rightarrow$ no. of distinct chars in A is as minimum as possible

Return the count of minimum distinct chars

Str = "a b f a d" B = 3

01. Need to know freq of each & every character

2	1	0	1	0	1	0
0	1	2	3	4	5	...25
a	b	c	d	e	f	z

AL $\Rightarrow \{2, 1, 1, 1\}$

* Sort this AL $\rightarrow \{1, 1, 1, 2\}$

B = ~~3~~ 2 + 0

str = "abcabca"

$$B = 3$$

3	2	2	0	0	0	0
0	1	2	3	4	5	...25
a	b	c	d	e	f	z

$$AL \rightarrow \{3, 2, 2\}$$

Sort the AL $\rightarrow \{2, 2, 3\}$ Ans = 2

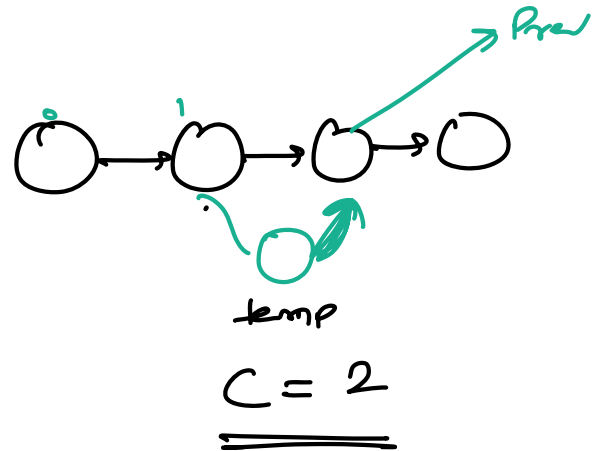
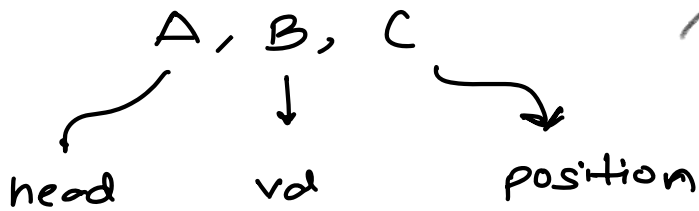
$$B = 3 - 2 = 1$$

*

$$A = \{ \underbrace{\text{"abab"}, \text{"abe"}, \text{"abcd"}} \}$$

comparison len = min(i' string, i+1' string)

Doubts



ListNode nn = (B)

prev ;

head = A

temp = head

i = 0 x 2

while (temp != null)

if (i == C)

prev = temp

temp = nn;

temp.next = prev;

;

temp = temp.next;

i++;

?

Motrix

res = [] []

add (matrix x)

res[res.length][j] + x[i][j]

* Contest → String, Hashing, Recursion